

Chapter: Perimeter and Area

1. Introduction

In geometry, we use two primary measurements to describe the size of 2D shapes:

- **Perimeter:** The total distance around the outside edge of a shape. Think of it as the length of a fence needed to enclose a yard.
 - **Area:** The total space inside the boundaries of a shape. Think of it as the amount of carpet needed to cover a floor or the amount of paint needed for a wall.
-

2. Perimeter

Definition: The perimeter is the total length of the boundary of a closed figure. For any polygon (a shape with straight sides), the perimeter is simply the sum of all its side lengths.

2.1 Perimeter of a Rectangle

A rectangle has two pairs of equal sides: the **Length (L)** and the **Width (W)**.

- **Formula:** $P = 2 \times (\text{Length} + \text{Width})$
- **Explanation:** Instead of adding $L + W + L + W$, we add the length and width once and multiply by two.

2.2 Perimeter of a Square

Since a square has four equal sides (s), we don't need to add them individually.

- **Formula:** $P = 4 \times \text{Side}$

2.3 Perimeter of an Equilateral Triangle

An equilateral triangle has three sides of equal length.

- **Formula:** $P = 3 \times \text{Side}$

2.4 Perimeter on a Grid

When measuring shapes on a grid:

1. **Straight segments:** Count the number of horizontal or vertical grid units.
 2. **Diagonal segments:** Note that diagonal lines across a grid square are longer than straight lines. In basic grid problems, you count the number of "diagonal units."
-

3. Area

Definition: Area is the measure of the region inside a closed shape, expressed in **square units** (e.g., cm^2 , m^2 , in^2).

3.1 Area of a Square

- **Formula:** $\text{Area} = \text{Side} \times \text{Side}$ (also written as s^2)

3.2 Area of a Rectangle

- **Formula:** $\text{Area} = \text{Length} \times \text{Width}$

3.3 Area of a Triangle

Every triangle can be seen as half of a rectangle or a parallelogram.

- **Formula:** $\text{Area} = \frac{1}{2} \times (\text{Base} \times \text{Height})$

3.4 Estimating Area of Irregular Shapes

To find the area of an irregular shape (like a handprint) on grid paper:

- **Full Square:** Count as **1** square unit.
- **More than half filled:** Count as **1** square unit.
- **Exactly half filled:** Count as **0.5** square unit.
- **Less than half filled:** Ignore (**0** square units).

4. Key Concept: Same Area, Different Perimeter

A vital concept in Grade 6 is that shapes with the same area do **not** necessarily have the same perimeter.

- **Compact Shapes:** Shapes that are "square-like" have the smallest perimeter for a given area.
- **Stretched Shapes:** Long, thin shapes have a much larger perimeter for the same area.

Example: If you have 9 square tiles (Area = 9 sq units):

1. **3x3 Square:** Perimeter = $4 \times 3 = 12$ units.
2. **9x1 Rectangle:** Perimeter = $2 \times (9 + 1) = 20$ units.

5. Summary Table

Shape	Perimeter Formula	Area Formula
Square	$4 \times s$	$s \times s$
Rectangle	$2 \times (L + W)$	$L \times W$
Triangle	Sum of all 3 sides	$\frac{1}{2} \times \text{Base} \times \text{Height}$

6. Solved Practice Problems

Problem 1: A wire in the shape of an equilateral triangle with a side of 12 cm is straightened and then bent into a square. What is the length of each side of the square?

- **Step 1:** Find total length of wire (Perimeter of triangle): $3 \times 12 = 36$ cm.
- **Step 2:** The perimeter of the square will also be 36 cm.
- **Step 3:** Side of square = $36 \div 4 = 9$ cm.

Problem 2: A rectangular sheet of paper is 15 cm long and 10 cm wide. A square of side 4 cm is cut out from the center. Find the remaining area of the sheet.

- **Step 1:** Area of original rectangle = $15 \times 10 = 150$ cm^2 .
- **Step 2:** Area of square cut-out = $4 \times 4 = 16$ cm^2 .
- **Step 3:** Remaining Area = $150 - 16 = 134$ cm^2 .